

MODULE *records*

Record constructors for the version 2 *Zcash P2P* protocol (draft *ZIP*, *zcash/zips#1344*): the *init* record of the handshake stream, announcement records, and the request/response formats of the *get-headers* and *get-blocks* request streams.

Blocks are abstracted to their height on a single linear chain, so a block hash, a block header and a block locator are all just heights.

Fields that are irrelevant to the properties being verified (nonce, services, user agent, relay preferences) are set to constant values to keep the *TLC* state space manageable, following the same convention as the legacy *messages.tla*.

EXTENDS *Naturals*, *Sequences*

The *init* record exchanged on the handshake stream (draft: “Init Record”).
version is the sender *s* advertised protocol version and *start_height*
 its best block height; both are the only fields the model acts on.

$\text{MakeInit}(\text{version}, \text{height}) \triangleq [$
 kind $\mapsto$ “init”,
 version $\mapsto$ *version*,
 services $\mapsto$ 0,
 nonce $\mapsto$ 0,
 user_agent $\mapsto$ “”,
 start_height $\mapsto$ *height*,
 relay $\mapsto$ 1,
 announce $\mapsto$ 0,
 full_ids $\mapsto$ 0
 $]$

A header announcement record on the block announcement stream
 (draft : “Block Announcements”, *kind* 0x00). The header is abstracted to
 the height it carries.

$\text{MakeHeaderAnnouncement}(\text{height}) \triangleq [$
 kind $\mapsto$ “header”,
 height $\mapsto$ *height*
 $]$

get-headers request (draft : “get-headers”). The locator is the requester *s*
 tip; *hash_stop* is zero (as many headers as possible); *tx_ids* is 0.

$\text{MakeGetHeaders}(\text{locator}) \triangleq [$
 kind $\mapsto$ “get-headers”,
 locator_count $\mapsto$ 1,
 locator $\mapsto$ *locator*,
 hash_stop $\mapsto$ 0,
 tx_ids $\mapsto$ 0
 $]$

get-headers response: the headers, as a sequence of heights.

$\text{MakeHeaders}(\text{heights}) \triangleq [$

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    kind    ↦ "headers",
    count   ↦ Len(heights),
    heights ↦ heights
]

get-blocks request (draft: "get-blocks"): the block hashes wanted.
MakeGetBlocks(heights) ≜ [
    kind    ↦ "get-blocks",
    count   ↦ Len(heights),
    heights ↦ heights
]

get-blocks response: the blocks delivered, a prefix of the request in
request order (the responder may finish after any complete entry).
MakeBlocks(heights) ≜ [
    kind    ↦ "blocks",
    count   ↦ Len(heights),
    heights ↦ heights
]

```
